PRACTICAL SESSION 10

Task 1: Investigate inode structures using commands such as ls -i, stat, and find. Create hard links and symbolic links and explain their impact on inode allocation.

```
(base) saiabburi@Charishma-MacBook-Air ~ % mkdir -p ~/Practical10
(base) saiabburi@Charishma-MacBook-Air ~ % cd ~/Practical10
(base) saiabburi@Charishma-MacBook-Air Practical10 % echo "This is the original file for inode practical." > original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % ls -i original.txt
68627237 original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % stat original.txt
16777230 68627237 -rw-r--r-- 1 saiabburi staff 0 47 "Oct 13 11:46:05 2026" "Oct 13 11:46:05 2026" "Oct 13 11:46:05 2026" "Oct 13 11:46:05 2026" 4096 8 0 original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % find . -name "original.txt"
./original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % ln original.txt hardlink.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % ls -i original.txt hardlink.txt
68627237 hardlink.txt   68627237 original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % stat -f "Inode: %i | Link count: %l | File: %N" original.txt
Inode: 68627237 | Link count: 2 | File: original.txt
```

```
(base) saiabburi@Charishma-MacBook-Air Practical10 % ln -s original.txt symlink.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % ls -li original.txt hardlink.txt symlink.txt
68627237 -rw-r--r--  2 saiabburi  staff  47 Oct 13 11:46 hardlink.txt
68627237 -rw-r--r--  2 saiabburi  staff  47 Oct 13 11:46 original.txt
68627269 lrwxr-xr-x  1 saiabburi  staff  12 Oct 13 11:46 symlink.txt -> original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % readlink symlink.txt
original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % original.txt
zsh: command not found: original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % echo "Added through hard link." >> hardlink.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % cat original.txt
This is the original file for inode practical.
Added through hard link.
(base) saiabburi@Charishma-MacBook-Air Practical10 % rm original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % cat hardlink.txt
'This is the original file for inode practical.
Added through hard link.
(base) saiabburi@Charishma-MacBook-Air Practical10 % ls -l symlink.txt
lrwxr-xr-x  1 saiabburi  staff  12 Oct 13 11:46 symlink.txt -> original.txt
(base) saiabburi@Charishma-MacBook-Air Practical10 % cat symlink.txt
cat: symlink.txt: No such file or directory
(base) saiabburi@Charishma-MacBook-Air Practical10 %
```

Task 2: Design a program using mmap() to perform file reading and writing. Compare memory-mapped I/O with traditional read/write operations in terms of efficiency and implementation complexity.

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/mman.h>
#include <sys/stat.h>

int main(void) {
    const char *filename = "mmap_data.txt";
    const char *message = "Updated using memory-mapped I/O.\n";

    int fd = open(filename, O_RDWR | O_CREAT, 0644);
    if (fd == -1) {
        perror("open");
        return 1;
    }

    /* Make the file large enough for the new content */
    size_t length = strlen(message);

    if (ftruncate(fd, (off_t)length) == -1) {
        perror("ftruncate");
        close(fd);
        return 1;
    }

    /* Map the file into the process address space */
    char *mapped = mmap(NULL, length,
                        PROT_READ | PROT_WRITE,
                        MAP_SHARED, fd, 0);

    if (mapped == MAP_FAILED) {
        perror("mmap");
        close(fd);
        return 1;
    }

    /* Write directly through the mapped memory */
    memcpy(mapped, message, length);

    if (msync(mapped, length, MS_SYNC) == -1) {
        perror("msync");
        munmap(mapped, length);
        close(fd);
        return 1;
    }
```

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    /* Unmap and close the file */
    if (munmap(mapped, length) == -1) {
        perror("munmap");
        close(fd);
        return 1;
    }

    if (close(fd) == -1) {
        perror("close");
        return 1;
    }

    /* Read back using traditional read() */
    fd = open(filename, O_RDONLY);
    if (fd == -1) {
        perror("reopen");
        return 1;
    }

    char buffer[256];
    ssize_t count = read(fd, buffer, sizeof(buffer));

    if (count < 0) {
        perror("read");
        close(fd);
        return 1;
    }

    printf("\nContent read using read():\n");
    if (count > 0) {
        fwrite(buffer, 1, (size_t)count, stdout);
    }

    close(fd);
    return 0;
}
```

```
(base) saiabburi@Charishma-MacBook-Air Practical10 % nano mmap_file.c
(base) saiabburi@Charishma-MacBook-Air Practical10 % clang -Wall -Wextra mmap_file.c -o mmap_file
(base) saiabburi@Charishma-MacBook-Air Practical10 % ./mmap_file
Content written using mmap():
Updated using memory-mapped I/O.

Content read using read():
Updated using memory-mapped I/O.
(base) saiabburi@Charishma-MacBook-Air Practical10 % cat mmap_data.txt
Updated using memory-mapped I/O.
(base) saiabburi@Charishma-MacBook-Air Practical10 %
```